

Joint Recovery: BARR Pair-Fusion

Frozen candidate B: problem model, analytical screening and algorithms

Fixed-contact model. Let V be a set of predefined visibility windows. A contact $v = (s_v, a_v, b_v, e_v)$ has a fixed satellite, antenna, start and end. The decision selects or omits the whole contact. Its weight is $w_v = e_v - b_v > 0$, represented by integer microsecond ticks in the stored graphs. On a shared resource r , order two contacts by start time. They conflict when the later start minus the earlier end is strictly less than the resource separation δ_r . Equality is compatible. The conflict graph $G = (V, E)$ combines satellite and antenna conflicts. The objective is

$$\max_{x \in \{0,1\}^{|V|}} \sum_{v \in V} w_v x_v, \quad x_u + x_v \leq 1 \quad \text{for each } \{u, v\} \in E. \quad (1)$$

For an independent set S , write $W(S) = \sum_{v \in S} w_v$.

Recovery relative to a fixed state. For each outsider $v \notin S$, its selected blockers are $B_v = N(v) \cap S$. For an independent outsider subset T , define

$$B(T) = \bigcup_{v \in T} B_v, \quad S_T = (S \setminus B(T)) \cup T, \quad g_S(T) = W(T) - W(B(T)). \quad (2)$$

Removing every selected blocker makes S_T feasible. The deficit of an outsider is $d_v = W(B_v) - w_v$; a positive singleton witness has $d_v < 0$. For compatible outsiders a, b , inclusion-exclusion gives

$$g_S(\{a, b\}) = W(B_a \cap B_b) - d_a - d_b, \quad (3)$$

$$U_{ab} = \min\{W(B_a), W(B_b)\} - d_a - d_b \geq g_S(\{a, b\}). \quad (4)$$

The scout rejects $U_{ab} \leq 0$ before computing the exact intersection. When all $d_v \geq 0$, a positive pair must share a selected blocker.

State, archive and budget quantities. The main search maintains a population $\mathcal{P}$ and an immutable historical best mask S^* . A call `remember( $S$ )` updates S^* only if its exact objective is larger. An event's state gain $g_S(T)$, an archive refresh $\max(0, W(S_T) - W(S^*))$, and a difference between budget-endpoint objectives have different reference states. The original B log stores state gain and refinement gain; it has no separate per-event archive-refresh field.

Initial states and local state. The saved common mask is the best of deterministic greedy orders using exponents $0, \frac{1}{2}, 1$. Member 0 descends from that mask; other members use diverse noisy greedy starts. Local state maintains the selected mask, blocker counts and blocker weights. Queue descent uses strict singleton improvements and one-to-many exchanges. Perturbations use a transaction; an interrupted or decreasing transaction is rolled back, while equality can be accepted.

Execution scope. The frozen B path is serial, with population 4 and one native thread. Native 360-second timing includes internal population construction; external input preparation and post-run validation are separate. The diagram expands the following three algorithms using matching numbers.

Population search and sparse fusion pulses

Algorithm 1: Frozen B pair-fusion population controller

Input : Fixed graph G, w , saved common mask S_0 , seed, budget $B = 360$ s
Output : Feasible historical incumbent S^*

- 1 Start the native clock; initialize population $\mathcal{P}$, S^* and exact-mask caches;
- 2 Set event count $e \leftarrow 0$, stale-round counter $k \leftarrow 0$ and charged time $H \leftarrow 0$;
- 3 **while** *global time remains* **do**
- 4 Advance each local member (at most 64 iterations or a 0.025 s slice);
- 5 Offer member states to remember; reset k on an integer-best refresh, else increment it;
- 6 Let t be elapsed time and $c = 0.05t - H$;
- 7 **if** $t < 30$ s, $k < 8$ rounds, *cooldown* < 0.5 s, or $c < 0.01$ s **then**
- 8 **continue**;
- 9 $e \leftarrow e + 1$; set event deadline to $\min(B, t + \min(c, 0.5 \text{ s}))$;
- 10 Select the elite on odd events; on even events rotate through members;
- 11 **if** $e \bmod 4 = 0$ and *event remaining time* > 0.04 s **then**
- 12 Choose a member farthest from the elite by weighted Hamming distance;
- 13 Fuse their union using component-wise bipartite min-cut (slice 0.1 s);
- 14 Descend the fused state; offer it to remember; use it as event state;
- 15 **if** the exact state mask has a cached complete zero certificate **then**
- 16 For a fused state, apply eligible population feedback;
- 17 **else**
- 18 $(T, g, \text{flags}) \leftarrow$ Algorithm 2 on the event state S ;
- 19 Cache S only when the flags supply a complete zero certificate;
- 20 **if** $g > 0$ **then**
- 21 Form S_T ; verify full-graph feasibility and the exact objective;
- 22 remember(S_T) ; // positive witness precedes refinement
- 23 $\hat{S} \leftarrow$ Algorithm 3(S, T, S_T , deadline);
- 24 Verify $\hat{S}$; offer it to remember; reset the local state and $k \leftarrow 0$;
- 25 For a fused state, replace the worst slot only when not decreasing it;
- 26 Charge the complete event duration to H ;
- 27 **return** S^* ;

Parent-union fusion. Common selected vertices are fixed. The symmetric difference of two independent parents is bipartite; a weighted min-cut gives a compatible selection on each component. A component fallback preserves the better parent contribution when bounded work stops. This pulse produces a temporary event state before the pair scout. Fused-state feedback and the event’s pair exchange are distinct operations.

Cache identity and timing. Zero-cache hits require bitwise equality of the whole mask under the fixed graph and weights. There are per-slot caches and a separate fused-state cache. Partial scans do not create zero certificates. The measured `escalation_seconds` is an umbrella that overlaps scout, fusion, construction and recovery times; these fields are not disjoint summands.

Shared-blocker scout and certificate scope

Algorithm 2: Analytical shared-blocker screening

Input : Feasible mask S , fixed G, w , event index e , event deadline

Output : Best retained singleton/pair witness, exact gain and completion flags

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1 For all outsiders compute  $d_v$ ; retain the best positive singleton;
2 Build sorted flat blocker lists  $B_v$ ; check maintained counts and weights;
3 Rotate the outsider seed order by  $9973e \bmod |V \setminus S|$ ;
4 foreach seed  $a$  in the all-outsider order do
5   foreach outsider  $b$  sharing a selected blocker with  $a$  do
6     Deduplicate with stamps; process only  $a < b$ ;
7     if the event deadline is reached then
8        $\perp$  mark partial; return retained witness and flags;
9     if  $U_{ab} \leq 0$  or  $\{a, b\} \in E$  then
10       $\perp$  continue;
11     Compute  $I = W(B_a \cap B_b)$  by a merge of sorted lists;
12     Set  $g = I - d_a - d_b$ ;
13     Retain the largest positive gain (fewer inserted vertices, then IDs, break ties);
14 If all seeds were covered and all  $d_v \geq 0$ , mark complete  $\leq 2$  coverage;
15 If additionally no positive witness exists, mark a complete zero certificate;
16 return retained witness, gain and flags;
```

Coverage argument. If a compatible pair has no shared blocker, then $g_S(\{a, b\}) = -d_a - d_b \leq 0$ when all singleton gains are nonpositive. Every possibly positive pair is therefore encountered by the selected blocker's two-hop adjacency traversal. The upper bound cannot reject a positive pair because the intersection weight is at most either blocker weight. Exact list intersection determines the gain of each surviving pair.

Complete and partial outcomes. The frozen setting `pair-max-seeds=0` requests all outsiders. The elapsed deadline can still stop enumeration. A partial scan provides its verified positive lower-bound witness, or an unknown result if none was found. Complete zero means only that this fixed state admits no improving exchange inserting at most two outsiders. It does not extend to three or more outsiders or to global optimality.

Three-outsider example. Let selected vertices u, v, z be independent, each of weight 3. Let three independent outsiders have weight 4 and blocker sets $\{u, v\}, \{v, z\}, \{z, u\}$. Every singleton gain is -2 , every pair gain is -1 , and the triple gain is $12 - 9 = 3$.

Cost of a scan. Flat blocker construction traverses selected-to-outsider adjacency. Pair intersections cost $O(|B_a| + |B_b|)$ after the cheap upper-bound and compatibility checks. Two-hop pair count depends on the actual blocker degrees. Neither all-seed configuration nor a zero certificate changes the event's fixed time allowance.

Positive-seed expansion and structured recovery

Algorithm 3: Positive-witness structured refinement

Input : Event state S , verified seed T , retained feasible S_T , deadline

Output : Retained feasible candidate $\hat{S}$

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1 Set  $\hat{S} \leftarrow S_T$ ; mark blockers  $B(T)$  as already paid;
2 Collect outsiders through paid blockers; rank by  $w_v - W(B_v \setminus B(T))$ ;
3 Expand to at most 12 outsiders, retaining candidates with negative singleton gain;
4 Compute a clique-cost upper bound  $U$  and a feasible sketch witness;
5 Verify and retain a sketch only if it exceeds the current witness;
6 if  $U > W(\hat{S}) - W(S)$  and time remains then
7   Induce outsiders and all affected backbone components with full boundary checks;
8   if the full Kernel fits the 4096-vertex limit then
9     Compile backbone response factors; attempt bounded-width elimination;
10    If unresolved and time remains, use bounded branch or feasible recovery;
11    Lift the result; verify it on  $G$ ; retain only an additional improvement;
12 On a deadline or size limit retain the already verified candidate;
13 return  $\hat{S}$ ;
```

Clique-cost bound. Partition the expanded outsiders into conflict cliques $\mathcal{C}$. For each blocker b , let h_b be the number of cliques adjacent to it. Assign $\lfloor w_b/h_b \rfloor$ to every adjacent outsider, and define

$$r_v = w_v - \sum_{b \in B_v} \lfloor w_b/h_b \rfloor, \quad U = \sum_{C \in \mathcal{C}} \max(0, \max_{v \in C} r_v). \quad (5)$$

A feasible subset takes at most one vertex per clique, so its allocated cost for b is at most $h_b \lfloor w_b/h_b \rfloor \leq w_b$. Thus U bounds the gain within this fixed expansion. The feasible sketch tries up to four lead orders and can pass through negative marginal prefixes while retaining the best positive subset.

Kernel response. For a general affected bipartite backbone component C , the response to outsider choice T is $\text{MWIS}(G[C \setminus N(T)])$. The pair path's backbone is the current independent set; each component is a singleton b with response $w_b \mathbf{1}[N(b) \cap T = \emptyset]$. Original outsider conflicts remain hard constraints. Exact flags apply to the bounded recovery subproblem when its bounds close or its structural computation completes.

Parameter	Frozen value	Parameter	Frozen value
Population / threads	4 / 1	Recovery slice	0.03 s
Branch-node limit	128	Elimination width / boundary	10 / 10
Factor-entry limit	262144	Expanded outsider cap	12
Event slice / fraction	0.5 s / 0.05	Fusion interval	every 4 gates

Source mapping.

Algorithm 1: `pair_solver_impl.hpp`.

Algorithm 2: `pair_scout.hpp`.

Algorithm 3: `selective.hpp`, `recombination.hpp`, `coupling.hpp`.

The 47 frozen B source hashes, input hashes, raw results and exact batch summaries are stored under `provenance/` and `results/`.

Historical B binary SHA256 prefix: `fb56e65a09f59762`.